MTH1003

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2018

Mathematical Modelling

Module Leader: Prof. J. Thuburn

Duration: 2 HOURS.

The mark for this module is calculated from 60% of the percentage mark for this paper plus 40% of the percentage mark for associated coursework.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) A ball is projected from the origin with speed V at an angle α to the horizontal. Neglecting air resistance and taking the gravitational acceleration to be g, write down expressions for the ball's horizontal and vertical coordinates x and z as functions of time t. Hence obtain the equation of the trajectory. Deduce the range of the ball as a function of V and α. (8)

   (b) Find all the equilibrium points with $x,\ y \in \mathbb{R}$ of the following dynamical system:

$$\dot{x} = (x^2 - y^2)x, \qquad \dot{y} = (y-2)(x^3 - 1).$$

   Classify the stability of the equilibrium point at $x = 0$, $y = 0$ for the following dynamical systems:

$$\text{(i)} \quad \dot{x} = x + 3y, \quad \dot{y} = -3x + y;$$
$$\text{(ii)} \quad \dot{x} = x + 3y, \quad \dot{y} = 3x + y.$$

   (12)

   (c) A particle of mass m, position x, moves in one dimension under a force given by a potential $V(x)$. Write down the equation of motion of the particle. Write down an expression for the total energy of the particle and show that it is conserved. (8)

   (d) Write down the forward Euler time stepping scheme for the solution of the ODE $dy/dt = f(y,t)$; use the notation y_n to indicate the approximate solution at time $t = nh$ where h is the time step size.

   Calculate the amplification factor for the scheme in the case $f(y,t) = -\nu y$ with ν a complex constant. Hence deduce the range of h for which the scheme is stable if $\nu = 2 + 2\mathrm{i}$. (12)

   (e) State Hooke's law for the tension in an elastic spring, defining all mathematical variables you use. A particle of mass m slides on a smooth horizontal table. The particle is attached to one end of a light elastic spring; the other end of the spring is fixed to a point on the table. Assuming the particle moves only in the direction of the spring, obtain its equation of motion. Calculate the frequency at which the system oscillates when displaced from its equilibrium position. (10)

[**50**]

SECTION B

2. You are given the equation of motion in polar coordinates (r, θ) for a particle of mass m moving under a central force $\mathbf{F}(r) = mf(r)\hat{\mathbf{r}}$:

$$(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right)\hat{\boldsymbol{\theta}} = f(r)\hat{\mathbf{r}},$$

where $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ are the unit vectors in the radial and azimuthal directions, respectively.

(a) Define the angular momentum per unit mass h, and show that it is conserved. (4)

(b) Eliminate $\dot{\theta}$ in terms of h, and hence re-express the radial equation of motion in terms of $u = 1/r$ with independent variable θ. (9)

(c) A particle is found to follow the path

$$r = \frac{r_0}{2\cos\theta - 1}$$

where r_0 is a constant. Determine the form of the force $f(r)$ governing the motion. (7)

(d) Explain why a force of the form found in part (c) cannot support steady circular orbits. (5)

[**25**]

3. (a) The following Matlab code is intended to use the midpoint rule to estimate a definite integral of the function given by a Matlab function `myf.m`. However, it contains a syntax error and an algorithmic error. Identify the two errors and explain how they should be fixed.

```
% Range and number of intervals
a = 0; b = 1; n = 20;

% Intervals
x = linspace(a,b,n-1);

% Contribution from each interval
while m = 1:n
    A(m) = (x(m+1) - x(m))*myf(0.5*(x(m) + x(m+1)));
end

% Total integral
integral = sum(A(:))
```

(6)

(b) Given two data points $(x_m, f(x_m))$, $(x_{m+1}, f(x_{m+1}))$, explain how the area under the curve $f(x)$ between x_m and x_{m+1} can be estimated as the area of a trapezium. Hence explain how the above Matlab script should be modified in order to implement the trapezium rule instead of the midpoint rule. (9)

(c) The equation

$$\dot{P} = aP - bP^2,$$

where a and b are positive constants, is intended to model the dynamics of a population of rabbits in the absence of predators. Give an interpretation of the two terms on the right hand side.

Find the solution of this equation given initial condition $P = P_0$. (10)

[**25**]

4. (a) A particle of mass $m = 1$ moves along the x-axis under a conservative force given by the potential

$$V(x) = x^2 \mathrm{e}^{-x^2/2}.$$

Find the force $F(x)$ associated with the potential.

Calculate $V'(x)$ and $V''(x)$. Hence find the positions of the equilibrium points, their stability, and the frequency of small oscillations about the positions of any stable equilibria. (10)

(b) Suppose the particle is projected from $x = 0$ with velocity $\dot{x} = U$. Write down the energy equation and find the total energy of the particle. Determine the ranges of U for which (i) $x \to \infty$ as $t \to \infty$, (ii) $x \to -\infty$ as $t \to \infty$, (iii) the particle oscillates. (8)

(c) A light elastic string has natural length l_0 and elastic modulus λ. The ends of the string are joined to form a loop, which is then wrapped around a cyclinder of radius l_0. Calculate the elastic potential energy in the string. (7)

[**25**]